

HE2002 Macroeconomics II

Tutorial 11

Academic Year 2025/2026, Semester 2

Quantitative Research Society @NTU

August 17, 2026

Overview

This tutorial covers:

- steady states in the Solow model with and without effective labour;
- per-worker and per-effective-worker transformations of production functions;
- steady-state output, capital, and consumption under alternative saving and depreciation rates;
- the consumption-maximising saving rate and its graphical interpretation;
- balanced-growth implications of technological progress and labour-force growth.

Contents

Question 1: Steady State and the Consumption-Maximising Saving Rate	3
Problem	3
Solution	4
Method 1: Direct steady-state algebra in per-worker form	4
Case $s = 0$	5
Case $s = 0.1$	5
Case $s = 0.2$	5
Case $s = 1$	5
Method 2: Golden-rule interpretation and break-even investment	7
Question 2: Steady State with Technological Progress and Population Growth	9
Problem	9
Solution	11
Method 1: Direct effective-worker steady-state algebra	11
Method 2: Balanced-growth-path interpretation	14

Question 1: Steady State and the Consumption-Maximising Saving Rate

Chapter 11, Q6.

Problem

Suppose that an economy is characterized by the following production function

$$Y = 3\sqrt{K}\sqrt{N}.$$

- (a) Derive the steady-state level of output per worker and the steady-state level of capital per worker in terms of the saving rate, s , and the depreciation rate, δ .
- (b) Derive the equation for the steady-state level of consumption per worker in terms of the saving rate, s , and the depreciation rate, δ .
- (c) Suppose that $\delta = 0.02$, compute the steady-state consumption per worker and steady-state output per worker when $s = 0$; $s = 0.1$; $s = 0.2$; $s = 1$. What do you observe?
- (d) Using your favorite spreadsheet software, plot your results on a graph showing the relationship between the saving rate (on the X axis) and consumption per worker (on the Y axis). Is there a saving rate that maximizes consumption per worker? Explain.

Solution

Method 1: Direct steady-state algebra in per-worker form

Let

$$k := \frac{K}{N}, \quad y := \frac{Y}{N}, \quad c := \frac{C}{N}.$$

Starting from

$$Y = 3\sqrt{K}\sqrt{N},$$

divide both sides by N :

$$\frac{Y}{N} = 3 \frac{\sqrt{K}\sqrt{N}}{N} = 3\sqrt{\frac{K}{N}} = 3\sqrt{k}.$$

Hence the per-worker production function is

$$y = 3\sqrt{k}.$$

With no population growth and no technological progress, the Solow law of motion for capital per worker is

$$k_{t+1} - k_t = sy_t - \delta k_t.$$

At the steady state,

$$k_{t+1} = k_t = k^*,$$

so

$$sy^* = \delta k^*.$$

Substituting $y^* = 3\sqrt{k^*}$,

$$3s\sqrt{k^*} = \delta k^*.$$

(a) Solve

$$3s\sqrt{k^*} = \delta k^*.$$

For the positive steady state, divide both sides by $\sqrt{k^*}$:

$$3s = \delta\sqrt{k^*}.$$

Hence

$$\sqrt{k^*} = \frac{3s}{\delta}, \quad k^* = \left(\frac{3s}{\delta}\right)^2.$$

Therefore the steady-state capital per worker is

$$\boxed{k^* = \left(\frac{3s}{\delta}\right)^2}.$$

The steady-state output per worker is

$$y^* = 3\sqrt{k^*} = 3\left(\frac{3s}{\delta}\right) = \frac{9s}{\delta}.$$

Therefore

$$\boxed{y^* = \frac{9s}{\delta}}.$$

(b) Steady-state consumption per worker is

$$c^* = (1 - s)y^*.$$

Using part (a),

$$c^* = (1 - s)\frac{9s}{\delta}.$$

Expanding,

$$c^* = \frac{9}{\delta}(s - s^2).$$

Thus

$$c^* = \frac{9s(1 - s)}{\delta} = \frac{9}{\delta}(s - s^2).$$

(c) Now set $\delta = 0.02$. Then

$$y^* = \frac{9s}{0.02} = 450s, \quad c^* = \frac{9s(1 - s)}{0.02} = 450s(1 - s).$$

Case $s = 0$.

$$y^* = 450(0) = 0, \quad c^* = 450(0)(1 - 0) = 0.$$

Case $s = 0.1$.

$$y^* = 450(0.1) = 45, \quad c^* = 450(0.1)(0.9) = 40.5.$$

Case $s = 0.2$.

$$y^* = 450(0.2) = 90, \quad c^* = 450(0.2)(0.8) = 72.$$

Case $s = 1$.

$$y^* = 450(1) = 450, \quad c^* = 450(1)(0) = 0.$$

Therefore,

s	0	0.1	0.2	1
y^*	0	45	90	450
c^*	0	40.5	72	0

We observe that steady-state output per worker rises with the saving rate, but steady-state consumption per worker does not. Consumption is zero when $s = 0$ and also zero when $s = 1$, so it must rise at first and then fall.

(d) Since

$$c^* = \frac{9}{\delta}(s - s^2),$$

the graph of steady-state consumption per worker against s is a downward-opening parabola.

To find the maximizing saving rate, differentiate:

$$\frac{dc^*}{ds} = \frac{9}{\delta}(1 - 2s).$$

Set this equal to zero:

$$1 - 2s = 0 \implies s = \frac{1}{2}.$$

The second derivative is

$$\frac{d^2c^*}{ds^2} = -\frac{18}{\delta} < 0,$$

so this indeed gives a maximum.

Thus the saving rate that maximizes steady-state consumption per worker is

$$s^{GR} = \frac{1}{2}.$$

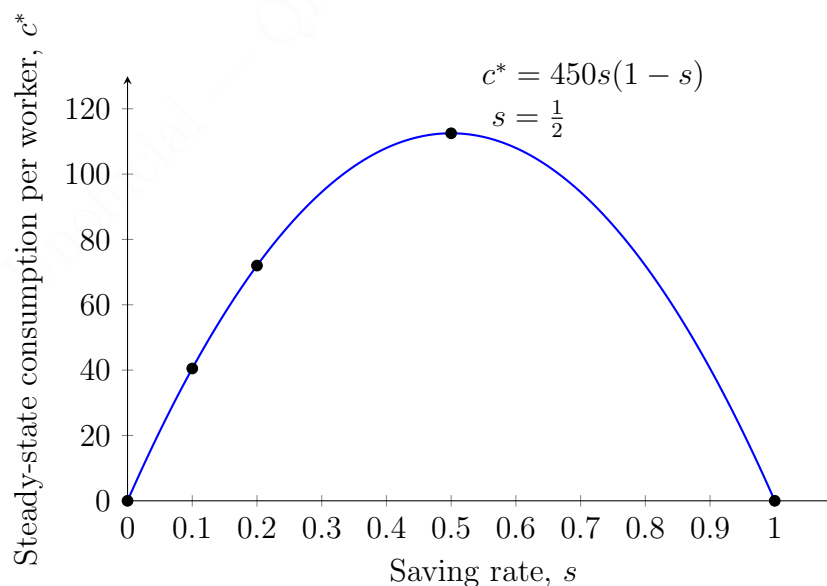
When $\delta = 0.02$, the maximum steady-state consumption per worker is

$$c_{\max}^* = \frac{9}{0.02} \left(\frac{1}{2} - \frac{1}{4} \right) = 450 \cdot \frac{1}{4} = 112.5.$$

Hence

$$c_{\max}^* = 112.5 \text{ when } \delta = 0.02.$$

A suitable graph is:



Therefore,

$$\text{Yes. The saving rate that maximizes steady-state consumption per worker is } s = \frac{1}{2}.$$

Method 2: Golden-rule interpretation and break-even investment

Again, the per-worker production function is

$$y = f(k) = 3\sqrt{k}.$$

In steady state,

$$sf(k) = \delta k.$$

Steady-state consumption per worker can also be written as

$$c^* = y^* - \delta k^*,$$

because at the steady state investment per worker equals depreciation per worker.

(a) Use

$$sf(k^*) = \delta k^*, \quad f(k) = 3\sqrt{k}.$$

Then

$$3s\sqrt{k^*} = \delta k^*.$$

As before,

$$\sqrt{k^*} = \frac{3s}{\delta}, \quad k^* = \left(\frac{3s}{\delta}\right)^2.$$

Hence

$$y^* = 3\sqrt{k^*} = 3 \cdot \frac{3s}{\delta} = \frac{9s}{\delta}.$$

Therefore

$$\boxed{k^* = \left(\frac{3s}{\delta}\right)^2, \quad y^* = \frac{9s}{\delta}.$$

(b) Using

$$c^* = f(k^*) - \delta k^*,$$

we obtain

$$c^* = 3\sqrt{k^*} - \delta k^*.$$

Substituting

$$\sqrt{k^*} = \frac{3s}{\delta}, \quad k^* = \left(\frac{3s}{\delta}\right)^2,$$

gives

$$c^* = 3\left(\frac{3s}{\delta}\right) - \delta\left(\frac{3s}{\delta}\right)^2 = \frac{9s}{\delta} - \frac{9s^2}{\delta} = \frac{9s(1-s)}{\delta}.$$

Hence

$$\boxed{c^* = \frac{9s(1-s)}{\delta}.$$

(c) With $\delta = 0.02$,

$$k^* = \left(\frac{3s}{0.02} \right)^2 = (150s)^2 = 22500s^2, \quad y^* = 450s, \quad c^* = 450s(1-s).$$

Therefore:

s	0	0.1	0.2	1
k^*	0	225	900	22500
y^*	0	45	90	450
c^*	0	40.5	72	0

The notable pattern is that higher saving always raises k^* and y^* , but not necessarily c^* . Very high saving can lower consumption because too much output is diverted into investment.

(d) A conceptually distinct way to find the consumption-maximizing saving rate is to maximize steady-state consumption with respect to k , not directly with respect to s . In the steady state,

$$c^* = f(k) - \delta k = 3\sqrt{k} - \delta k.$$

Differentiate with respect to k :

$$\frac{dc^*}{dk} = \frac{3}{2\sqrt{k}} - \delta.$$

Set equal to zero:

$$\frac{3}{2\sqrt{k}} = \delta \implies \sqrt{k} = \frac{3}{2\delta}.$$

Thus the golden-rule capital stock is

$$k^{GR} = \left(\frac{3}{2\delta} \right)^2.$$

At a steady state, the saving rate that supports a given k satisfies

$$s = \frac{\delta k}{f(k)} = \frac{\delta k}{3\sqrt{k}} = \frac{\delta\sqrt{k}}{3}.$$

Substituting $\sqrt{k^{GR}} = \frac{3}{2\delta}$,

$$s^{GR} = \frac{\delta}{3} \cdot \frac{3}{2\delta} = \frac{1}{2}.$$

Hence

$$\boxed{s^{GR} = \frac{1}{2}.$$

So there is indeed a unique saving rate that maximizes steady-state consumption per worker. It is the rate that supports the golden-rule capital stock.

Question 2: Steady State with Technological Progress and Population Growth

Chapter 12, Q6

Problem

Suppose that the economy's production function is

$$Y = K^{1/2}(AN)^{1/2},$$

that the saving rate, s , is equal to 16%, and that the rate of depreciation, δ , is equal to 10%. Suppose further that the number of workers grows at 2% per year and that the rate of technological progress is 4% per year.

To solve for the steady-state values of capital stock per effective worker $\left(\frac{K}{AN}\right)^*$ and output per effective worker $\left(\frac{Y}{AN}\right)^*$, we use the condition that actual investment I equals required investment I_R . This allows us to determine the steady-state level of capital per effective worker $\left(\frac{K}{AN}\right)^*$.

In our model with technological progress:

$$I = sY, \quad I_R = (\delta + g_A + g_N)K$$

$$I = I_R \Rightarrow sY = (\delta + g_A + g_N)K \Rightarrow s \frac{Y}{AN} = (\delta + g_A + g_N) \frac{K}{AN} \Rightarrow sf\left(\frac{K}{AN}\right) = (\delta + g_A + g_N) \frac{K}{AN}$$

The value of capital per effective worker that satisfies the equation above is the steady-state level of capital stock per effective worker, denoted as $\left(\frac{K}{AN}\right)^*$.

- (a) Find the steady-state values of the variables listed in (1) through (5).
 - i. The capital stock per effective worker.
 - ii. Output per effective worker.
 - iii. The growth rate of output per effective worker.
 - iv. The growth rate of output per worker.
 - v. The growth rate of output.
- (b) Suppose that the rate of technological progress doubles to 8% per year. Recompute the answers to part a.
 - i. The capital stock per effective worker.
 - ii. Output per effective worker.
 - iii. The growth rate of output per effective worker.
 - iv. The growth rate of output per worker.
 - v. The growth rate of output.

- (c) Now suppose that the rate of technological progress is still equal to 4%, but the number of workers now grows at 6% per year. Recompute the answers to part a. Are people better off in part a or part c? Explain.
- i. The capital stock per effective worker.
 - ii. Output per effective worker.
 - iii. The growth rate of output per effective worker.
 - iv. The growth rate of output per worker.
 - v. The growth rate of output.

Solution

Method 1: Direct effective-worker steady-state algebra

Define capital per effective worker and output per effective worker by

$$\tilde{k} := \frac{K}{AN}, \quad \tilde{y} := \frac{Y}{AN}.$$

Given

$$Y = K^{1/2}(AN)^{1/2},$$

divide by AN :

$$\tilde{y} = \frac{K^{1/2}(AN)^{1/2}}{AN} = \left(\frac{K}{AN}\right)^{1/2} = \tilde{k}^{1/2}.$$

So the per-effective-worker production function is

$$\tilde{y} = f(\tilde{k}) = \sqrt{\tilde{k}}.$$

In steady state with technological progress and labour-force growth,

$$sf(\tilde{k}^*) = (\delta + g_A + g_N)\tilde{k}^*.$$

With $f(\tilde{k}) = \sqrt{\tilde{k}}$, this becomes

$$s\sqrt{\tilde{k}^*} = (\delta + g_A + g_N)\tilde{k}^*.$$

For the positive steady state,

$$\sqrt{\tilde{k}^*} = \frac{s}{\delta + g_A + g_N}, \quad \tilde{k}^* = \left(\frac{s}{\delta + g_A + g_N}\right)^2,$$

and

$$\tilde{y}^* = \sqrt{\tilde{k}^*} = \frac{s}{\delta + g_A + g_N}.$$

(a) Here

$$s = 0.16, \quad \delta = 0.10, \quad g_A = 0.04, \quad g_N = 0.02.$$

Therefore

$$\delta + g_A + g_N = 0.10 + 0.04 + 0.02 = 0.16.$$

So

$$\tilde{k}^* = \left(\frac{0.16}{0.16}\right)^2 = 1, \quad \tilde{y}^* = \frac{0.16}{0.16} = 1.$$

In steady state, output per effective worker is constant, so its growth rate is

$$g_{\tilde{y}} = 0.$$

Output per worker is

$$\frac{Y}{N} = A \cdot \frac{Y}{AN} = A\tilde{y},$$

so in steady state its growth rate is

$$g_{Y/N} = g_A + g_{\tilde{y}} = g_A = 4\%.$$

Total output is

$$Y = (AN)\tilde{y},$$

so its growth rate is

$$g_Y = g_A + g_N + g_{\tilde{y}} = g_A + g_N = 6\%.$$

Therefore,

i. $\left(\frac{K}{AN}\right)^* = 1,$ ii. $\left(\frac{Y}{AN}\right)^* = 1,$ iii. $g_{Y/(AN)} = 0,$ iv. $g_{Y/N} = 4\%,$ v. $g_Y = 6\%.$
--

(b) Now let technological progress double to 8%, so

$$g_A = 0.08.$$

Then

$$\delta + g_A + g_N = 0.10 + 0.08 + 0.02 = 0.20.$$

Hence

$$\tilde{k}^* = \left(\frac{0.16}{0.20}\right)^2 = (0.8)^2 = 0.64, \quad \tilde{y}^* = \frac{0.16}{0.20} = 0.8.$$

Again, in steady state,

$$g_{Y/(AN)} = 0.$$

Output per worker grows at

$$g_{Y/N} = g_A = 8\%.$$

Output grows at

$$g_Y = g_A + g_N = 8\% + 2\% = 10\%.$$

Therefore,

i. $\left(\frac{K}{AN}\right)^* = 0.64,$ ii. $\left(\frac{Y}{AN}\right)^* = 0.8,$ iii. $g_{Y/(AN)} = 0,$ iv. $g_{Y/N} = 8\%,$ v. $g_Y = 10\%.$
--

(c) Now return to $g_A = 0.04$, but let the growth rate of workers rise to

$$g_N = 0.06.$$

Then

$$\delta + g_A + g_N = 0.10 + 0.04 + 0.06 = 0.20.$$

Therefore

$$\tilde{k}^* = \left(\frac{0.16}{0.20} \right)^2 = 0.64, \quad \tilde{y}^* = \frac{0.16}{0.20} = 0.8.$$

In steady state,

$$g_{Y/(AN)} = 0.$$

Output per worker grows at

$$g_{Y/N} = g_A = 4\%.$$

Output grows at

$$g_Y = g_A + g_N = 4\% + 6\% = 10\%.$$

Hence

i. $\left(\frac{K}{AN} \right)^* = 0.64,$ ii. $\left(\frac{Y}{AN} \right)^* = 0.8,$ iii. $g_{Y/(AN)} = 0,$ iv. $g_{Y/N} = 4\%,$ v. $g_Y = 10\%.$
--

People are better off in part a than in part c, because the steady-state level of output per effective worker is higher in part a:

$$1 > 0.8.$$

Also, since g_A is the same in parts a and c, the growth rate of output per worker is the same, namely 4%, but the level of output per worker is lower in part c because $\tilde{y}^*$ is lower.

Therefore,

People are better off in part a than in part c.

Method 2: Balanced-growth-path interpretation

A conceptually different way to proceed is to start from the balanced-growth properties of the Solow model with effective labour.

From

$$Y = K^{1/2}(AN)^{1/2},$$

if we divide by AN , we obtain

$$\frac{Y}{AN} = \left(\frac{K}{AN} \right)^{1/2}.$$

Thus constant $\frac{K}{AN}$ implies constant $\frac{Y}{AN}$. So on the balanced-growth path:

- capital per effective worker is constant;
- output per effective worker is constant;
- output per worker grows at the rate of technological progress g_A ;
- total output grows at the rate $g_A + g_N$.

To pin down the constant level of $\tilde{k}$, use the steady-state investment condition:

$$s\tilde{y} = (\delta + g_A + g_N)\tilde{k}.$$

Since $\tilde{y} = \sqrt{\tilde{k}}$, this gives

$$s\sqrt{\tilde{k}} = (\delta + g_A + g_N)\tilde{k},$$

hence

$$\sqrt{\tilde{k}^*} = \frac{s}{\delta + g_A + g_N}, \quad \tilde{k}^* = \left(\frac{s}{\delta + g_A + g_N} \right)^2, \quad \tilde{y}^* = \frac{s}{\delta + g_A + g_N}.$$

(a) Since

$$\delta + g_A + g_N = 0.10 + 0.04 + 0.02 = 0.16,$$

we get

$$\tilde{k}^* = 1, \quad \tilde{y}^* = 1.$$

Balanced-growth implications then give

$$g_{Y/(AN)} = 0, \quad g_{Y/N} = g_A = 4\%, \quad g_Y = g_A + g_N = 6\%.$$

Therefore,

$$\boxed{\begin{aligned} \left(\frac{K}{AN} \right)^* &= 1, \\ \left(\frac{Y}{AN} \right)^* &= 1, \\ g_{Y/(AN)} &= 0, \\ g_{Y/N} &= 4\%, \\ g_Y &= 6\%. \end{aligned}}$$

(b) If g_A doubles to 8%, then

$$\delta + g_A + g_N = 0.10 + 0.08 + 0.02 = 0.20.$$

So

$$\tilde{k}^* = \left(\frac{0.16}{0.20} \right)^2 = 0.64, \quad \tilde{y}^* = \frac{0.16}{0.20} = 0.8.$$

Since effective-worker variables are constant on the balanced-growth path,

$$g_{Y/(AN)} = 0.$$

Meanwhile,

$$g_{Y/N} = g_A = 8\%, \quad g_Y = g_A + g_N = 10\%.$$

Hence

$$\begin{aligned} \left(\frac{K}{AN} \right)^* &= 0.64, \\ \left(\frac{Y}{AN} \right)^* &= 0.8, \\ g_{Y/(AN)} &= 0, \\ g_{Y/N} &= 8\%, \\ g_Y &= 10\%. \end{aligned}$$

(c) If $g_A = 4\%$ and $g_N = 6\%$, then once again

$$\delta + g_A + g_N = 0.10 + 0.04 + 0.06 = 0.20.$$

So the steady-state effective-worker levels are the same as in part b:

$$\tilde{k}^* = 0.64, \quad \tilde{y}^* = 0.8.$$

But the growth decomposition changes:

$$g_{Y/(AN)} = 0, \quad g_{Y/N} = g_A = 4\%, \quad g_Y = g_A + g_N = 10\%.$$

Therefore,

$$\begin{aligned} \left(\frac{K}{AN} \right)^* &= 0.64, \\ \left(\frac{Y}{AN} \right)^* &= 0.8, \\ g_{Y/(AN)} &= 0, \\ g_{Y/N} &= 4\%, \\ g_Y &= 10\%. \end{aligned}$$

Comparing parts a and c:

- $\tilde{y}^*$ is higher in part a (1) than in part c (0.8);
- output per worker grows at the same rate in both parts, because g_A is unchanged at 4%;
- therefore the relevant welfare comparison is about the level of output per worker, which is higher in part a.

Thus

Part a is better than part c because the steady-state level of output per worker is higher.

Case	$\left(\frac{K}{AN}\right)^*$	$\left(\frac{Y}{AN}\right)^*$	$g_{Y/(AN)}$	$g_{Y/N}$	g_Y
(a) $g_A = 4\%$, $g_N = 2\%$	1	1	0	4%	6%
(b) $g_A = 8\%$, $g_N = 2\%$	0.64	0.8	0	8%	10%
(c) $g_A = 4\%$, $g_N = 6\%$	0.64	0.8	0	4%	10%